

System Overview, Conventions & Hyperparameter Atlas

What the Vol-Fitter does, the pipeline, and the master index of every knob
Vol-Fitter Technical Notes, No. 00

Vol-Fitter

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Abstract

This is the map of the territory. The Vol-Fitter is an implied-volatility surface fitter in the spirit of a professional vola-fitter, with one differentiating capability: it *extrapolates* sparse smile observations to a full universe of smiles, across expiries *and* assets, by propagating signal through a graph whose nodes are smiles (underlying, T). This note states the market and normalization conventions shared by the whole series, walks the end-to-end compute pipeline (fetch $\rightarrow$ forwards/carry $\rightarrow$ prepare [de-Americanize, screen, weight] $\rightarrow$ fit $\rightarrow$ views + graph), catalogues what the application can do, and ends with an index of the *principal calibration controls*, cross-linked to the topic notes that derive each knob. One claim organizes the whole series: every guarantee is graded and *executable* — structural where a model can carry it, certified at the publish gate where it cannot, measured everywhere else — and the production incidents that taught each design rule are frozen as named certification cases that re-run on demand. It is the front door: read it once, then any other note in the series stands alone.

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1 What the application is

At its core the Vol-Fitter solves one problem repeatedly — *given noisy, sparse option quotes for one expiry of one underlying, produce a smooth implied-volatility smile under explicit no-arbitrage control* — and then layers three ambitions on top of it:

1. **A choice of models, all arbitrage-aware.** Three slice parametrizations share the per-expiry calibration spine — the *LQD* log-quantile-density model (Note 01), *SVI* / *SVI-JW* (Note 02), and the *Multi-Core Sigmoid (MCS)* family (Note 03) — and a fourth model stands apart: the *piecewise-affine local-volatility surface* (Note 04), a *jointly* calibrated maturity $\times$ strike grid whose unknown is piecewise-affine local *variance*, priced through the Dupire PDE. On arbitrage the honest statement is graded — structural, certified, measured. LQD is butterfly-free *by construction* and the LV surface is structurally arbitrage-clean (positive local variance); SVI and MCS are *soft-fenced* by sampled penalties, which reduce but do not guarantee zero residual violation; and the grading no longer stops there: every *displayed* slice, whatever its model, must pass a dense-grid belly *butterfly certificate* over its traded range — Durrleman $g \geq -10^{-4}$ on an 801-point grid, computed from the model's own derivatives in a few dozen microseconds — before it counts as publishable; an uncertified slice cannot become a mark (Note 02). Residual violation rates are still measured and reported per fit, never assumed away.
2. **A professional desk workflow.** Real bid/ask handling (fit to mid, to the bid–ask band, or to a haircut band), var-swap targets, density-corrected weighting, event time dilation, dividend and forward treatment, spot–vol dynamics, a Bayesian prior-persistence machine that decides, feature by feature (operator by operator), whether the live quotes identify a smile feature or the saved prior should carry it, and a temporal observation filter (a per-handle Kalman/MAP estimator, Note 15) that denoises noisy or contradictory quotes while treating a genuine data gap as absent evidence rather than noise. The filter has three modes: **off** (byte-identical to the feature never existing), **overlay** (post-fit state and display update only), and **active** (the prediction enters the calibration objective before optimization); **active** is off by default and still carries a documented shock-day lag.
3. **Graph smile-extrapolation.** The headline differentiator (Note 14): a transported prior plus the lit-node calibration innovation are propagated through a *directed Bayesian graph* to reconstruct the *dark* nodes. The propagation operator is itself a menu — the pairwise *precision-message* operator (the options default since 2026-07-27), the **smooth_field** increment prior (the wire default and explicit rollback), and an opt-in layered dynamic-harmonic pipeline — with an optional optimal-transport regularization, off by default ($\lambda = 0$). The defaults here were chosen by held-out adjudication, not preference (section 4).

Heuristic

The unifying idea is that a volatility surface is not a curve to be drawn but a *family of probability laws* to be inferred — one risk-neutral law per (underlying, T) node — subject to no-arbitrage couplings (butterfly within a slice, calendar across expiries) and to a Bayesian prior that ties the whole universe together. Every design choice in the app follows from taking that sentence literally.

Vocabulary. Terms the whole series uses without ceremony, defined once. A *smile* is the implied-volatility curve of one (underlying, T); a *slice* is one smile’s fitted model; a *surface* is all expiries of one underlying; a *node* is one (underlying, T) pair viewed as a graph vertex. A node is *lit* when its quotes are consumed and calibrated, and *dark* otherwise — operationally, dark means the quotes are absent, unusable, deliberately withheld, or simply not consumed; it does not necessarily mean nobody quoted. The *prior* is the saved previous fit, transported to today’s market; the *innovation* is a lit node’s fitted-minus-prior move. A *handle* is a scalar smile summary — the three ATM handles are level σ_0 , skew s_0 and curvature κ_0 . The *haircut band* is the bid–ask band shrunk by a fixed vol amount; a *vol bp* is 0.01 vol point (10^{-4} in volatility); a *WW smile* is a W-shaped smile (central trough with two shoulders, Note 03). Abbreviations: CRR (Cox–Ross–Rubinstein binomial tree), SSR (skew-stickiness ratio, Note 12), OT (optimal transport), MAP (maximum a posteriori), GN (Gauss–Newton), NBBO (national best bid/offer).

Invariant

The system-wide contracts every note protects, stated once:

1. **Arbitrage-cleanliness is structural where possible, certified at the gate, measured always.** LQD and the LV surface are clean by construction; SVI/MCS are fenced by penalties that are exactly zero on admissible slices — which bounds but does not eliminate residual violation; the belly butterfly certificate hard-blocks readiness and publish for *any* uncertified displayed slice; and the residual rates are reported by explicit per-fit diagnostics, never assumed away.
2. **Every feature is strictly additive.** Bands, var-swap targets, calendar coupling, the event clock, priors, betas, the wing penalty, the observation filter: each is byte-identical to its absence when off — and that byte-identity is test-locked, not asserted.
3. **Numbers are measured, not copied.** Figures and tables are generated fresh by the production code; reference listings are executed against the modules they distill; claims carry test anchors.
4. **Versioned invalidation is scoped.** Editing one name’s rate, dividends, events or prior bumps that ticker’s version only — never the universe. Whether the name then *refits* immediately depends on the workflow gate: with `autoCalibrate` on it refits in the background; on the trigger-gated live server (which defaults `autoCalibrate` off) the existing fit is kept, marked stale, and waits for an explicit Calibrate.

Figure 1 shows the two static-arbitrage checks, each in its correct coordinates, on a four-expiry synthetic surface fitted by the production LQD calibrator. Panel (b) is the *butterfly* check: every implied density is non-negative, by construction. Panel (c) is the *calendar* check: stacked *total variance* $w(k, T)$, where for forward-normalized (mean-one) slices absence of calendar arbitrage at fixed log-moneyness is exactly the non-crossing ordering $w(k, T_{i+1}) \geq w(k, T_i)$ at *every* k — the same view the app’s Stacked IV tab draws. Two tempting substitutes are not checks at all: stacked *vol* smiles carry no calendar content (σ mixes w with the clock, so clean surfaces cross freely in vol), and ATM-only monotonicity is necessary but far from sufficient. The four slices here are fitted independently, so the displayed dominance is *measured* — the generator asserts a positive minimum inter-expiry gap on the displayed grid before writing the figure — not enforced; the enforcement

coupling is Note 10's calendar machinery.

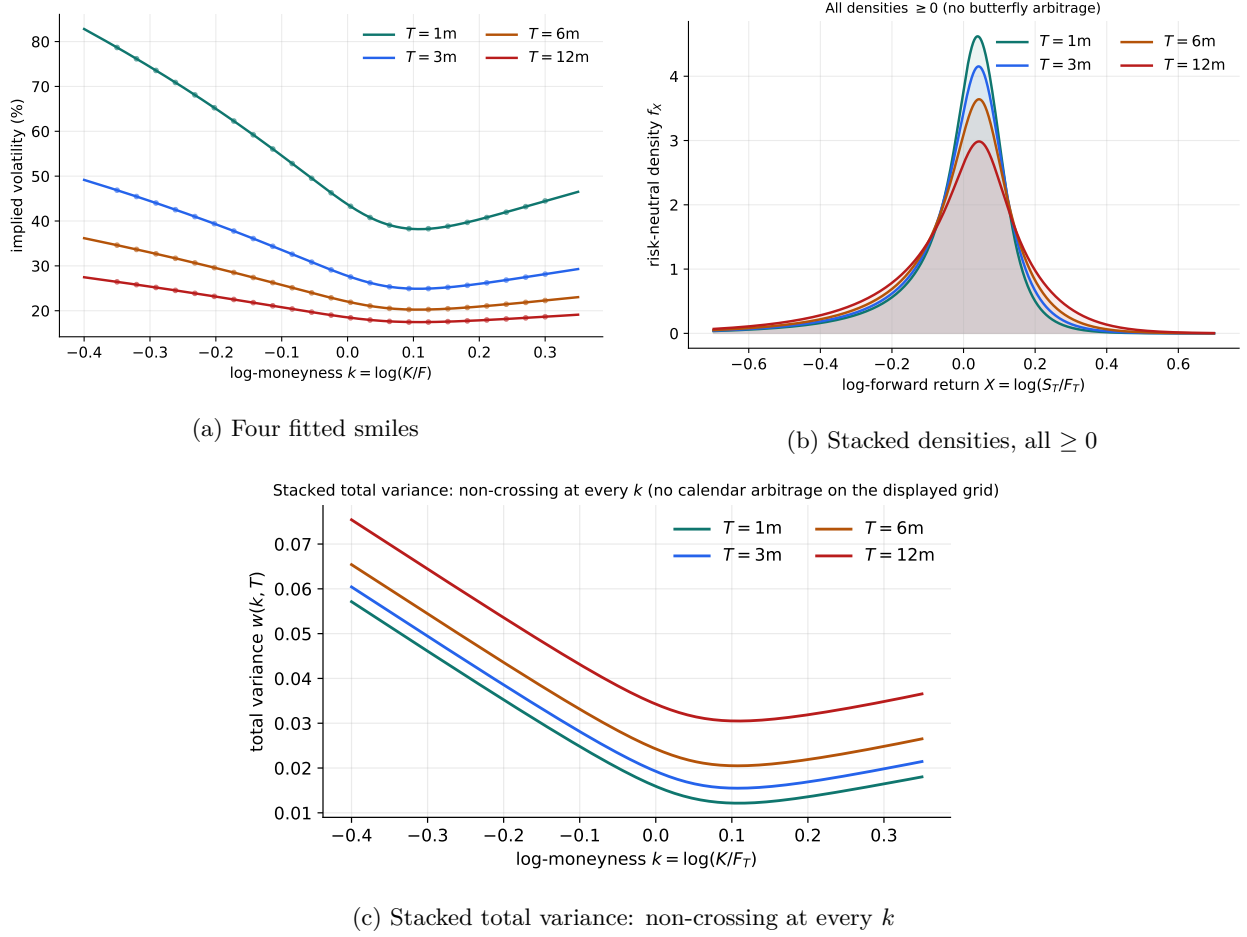


Figure 1: The two static-arbitrage checks, produced end to end by the production LQD fitter on a synthetic calendar-monotone surface. Panels a, b: the fits and their non-negative densities (no butterfly arbitrage, by construction). Panel c: stacked total variance, non-crossing at every displayed k (no calendar arbitrage there — measured on independently fitted slices; enforcement is Note 10's coupling).

2 Market conventions and normalization

These conventions hold across the entire series; individual notes assume them without restating.

Definition 1 (Forward measure and normalization). For expiry T with forward price F_T , work under the T -forward measure and in forward-normalized, undiscounted units. The log-moneyness is $k = \log(K/F_T)$, the log-forward return is $X_T = \log(S_T/F_T)$, and normalized call/put prices are $C_T(k) = \mathbb{E}^T[(e^{X_T} - e^k)^+]$ and $P_T(k) = \mathbb{E}^T[(e^k - e^{X_T})^+]$, with the martingale condition $\mathbb{E}^T[e^{X_T}] = 1$.

Definition 2 (Total variance and the Black map). The normalized Black call is $B(k, w) = \Phi(d_+) - e^k \Phi(d_-)$ with $d_{\pm} = -k/\sqrt{w} \pm \sqrt{w}/2$. Total implied variance $w(k, T)$ is the unique w solving $B(k, w) = C_T(k)$ — a property of the *price*, independent of any time convention; the fitter inverts it

by a vectorized scheme rather than a scalar root-find per strike. Annualizing w by a clock gives an implied volatility: $\sigma^2 = w/t$ under the calendar clock, $\sigma^2 = w/\tau$ under the event clock below.

This one primitive underlies every model in the series; in normalized form it is a handful of lines (matches production to 10^{-12}):

Listing 1: The normalized Black call $B(k, w)$ — the shared pricing primitive (distilled from `core/black.py`).

```

1 def black_call(k, w):
2     # k = log(K/F); w = total implied variance
3     sq = np.sqrt(w)
4     dp = -k/sq + 0.5*sq          # d_+; d_- = d_+ - sqrt(w)
5     price = norm_cdf(dp) - np.exp(k)*norm_cdf(dp - sq)
6     intrinsic = np.maximum(1 - np.exp(k), 0.0)
7     return np.where(w > 1e-12, price, intrinsic)

```

Three conventions deserve emphasis because they recur:

- **Two clocks, one w .** Four objects, kept distinct: t is the calendar year-fraction to expiry; w is the clock-independent total variance inverted from the price; $\sqrt{w/t}$ is the calendar-annualized market IV; $\sqrt{w/\tau}$ is the *working* (event-time) IV, where τ is calendar time dilated by scheduled events (Note 11). Fits live on τ when the event clock is on. “Adding an event lowers IV” means precisely: at fixed price (hence fixed w), the event raises τ and so lowers the *working* IV — the market IV and the price are untouched. De-Americanization and carry always run on the calendar clock t .
- **Forward-normalized everything.** Prices, strikes and densities are quoted relative to F_T , so a single smile object is spot-independent until a dynamics rule (Note 12) transports it under a spot move.
- **Residual units are per-model.** LQD and the LV surface calibrate on price residuals divided by Black vega — a volatility-like objective whose every iterate is a valid price (Notes 01, 04, 07); SVI and MCS fit directly in implied-volatility units. The bid-ask/hairecut band objective (Note 07) is expressed in vol space for all models.

3 The compute pipeline

Every smile the app shows is the output of the pipeline in fig. 2. The stages are deliberately decoupled and independently cached so that, e.g., editing a fit penalty never re-pulls data or re-runs de-Americanization.

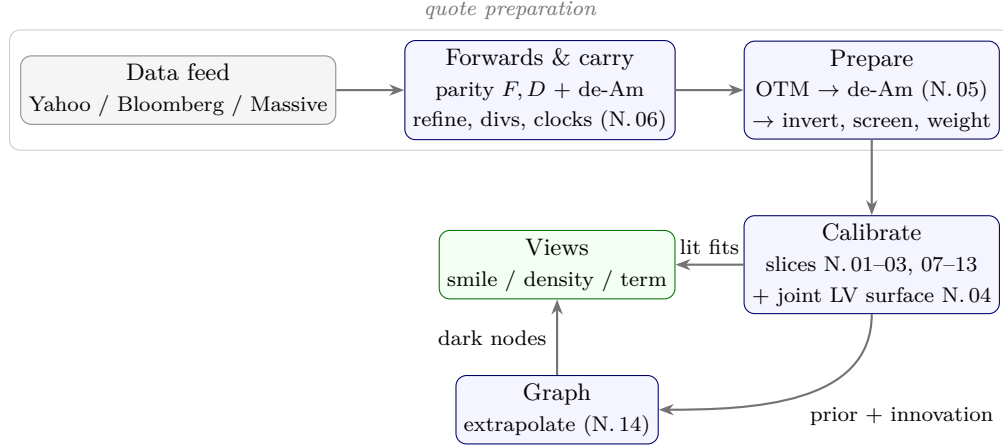


Figure 2: The end-to-end compute spine. The forward, discount, dividends and clocks are resolved first (parity with an American de-bias refinement); quote preparation then selects the OTM side, de-Americanizes each quote, and inverts/screens/weights; calibration (per-expiry slices plus the jointly fitted LV surface) feeds the lit views *and* the graph extrapolator, whose dark-node reconstructions feed the same views.

Stage by stage.

1. **Data feed.** A provider registry exposes Yahoo (US options), Bloomberg (xbbg; US + non-US indices/equities), and Massive (Polygon/OPRA), plus a synthetic source; an as-of selector serves Live / previous-close / EOD / captured-replay. US-options-only providers and the non-US Bloomberg path are handled transparently.
2. **Forwards, dividends and clocks** (Note 06). Resolved *before* any per-quote work, because everything downstream consumes them. Put-call parity on the raw chain yields the implied forward and discount (clamped against feed noise); the forward inference itself contains a near-ATM American de-bias loop, so the regressed forward is consistent with American quotes. Discrete dividends and the two clocks — calendar t and event-dilated variance time τ — are resolved here too. One provider pathology is handled upstream of the regression: chains a delayed feed *synthesizes* at zero carry (NBBO gated, so the provider emits Black prices from its own IV marks at $F = \text{spot}$, $D = 1$) are flagged at capture and *pinned* to $F = \text{spot}$, $D = 1$ rather than parity-regressed — regressing quotes that already embed the provider’s carry produced garbage forwards.
3. **Prepare** (Notes 05, 07). Per-expiry, given the resolved forward and clocks: select the out-of-the-money side of each strike; then *de-Americanize* every kept quote (Note 05) — listed equity options are American, and each is converted to its European-equivalent implied volatility by inverting a Cox–Ross–Rubinstein tree (bisection), with a Numba kernel for wide chains and a content-digest cache so fit-tuning never re-runs it; then normalize, invert to total variance (vectorized), apply the static no-arbitrage/wing screen — every dropped quote is quarantined with a reason, never silently discarded — run the wing-only, band-constrained convexity repair (holding the ATM core fixed), and assign observation weights (Note 07).
4. **Calibrate.** The three parametric families fit per-expiry slices; the LV surface is calibrated *jointly* across maturities and strikes from the same prepared quotes (Note 04). Each fit runs

under the active objective: mid / bid-ask band / haircut (Note 07), optional var-swap target (Note 08), calendar coupling (Note 10), event clock (Note 11), and Bayesian prior persistence (Note 13). The observation Kalman filter (Note 15) sequences by mode: `off` adds nothing (byte-identical); `overlay` updates the filtered state and display *after* the fit commits; `active` inserts the prediction prior into the calibration objective *before* optimization.

5. **Views.** Every smile-derived view — smile, stacked densities, stacked total variance, quantile density, term structure, local-vol surface, SSR scenario — follows the chosen model and is downsampled for the UI. Views serve lit fits and, once the graph has run, the reconstructed dark nodes.
6. **Graph** (Note 14). The transported prior plus the lit-node innovation propagate through the directed Bayesian graph (OT term optional, off by default) to reconstruct dark-node smiles with credible bands, which flow back into the views.

4 The smile graph

The object that makes the app more than a slice fitter is the *smile graph* (fig. 3): nodes are smiles (underlying, T); edges encode that one node informs another. Calendar edges link expiries of one name; cross-asset edges link an index or sector ETF to its single names. A user marks nodes *lit* (quoted, calibrated) or *dark* (to be reconstructed). The graph carries the three ATM handles (σ_0, s_0, κ_0) of each node as a Gaussian signal and propagates the lit innovation to the dark nodes. Three scalars do not make a smile: a dark node’s full curve is obtained by *retargeting* — the node’s transported prior smile (a complete curve, wings and all) is re-expressed in an ATM-orthogonal chart and its three handles are moved to the posterior values, leaving the prior’s shape modes (wings, event convexity) untouched, so every reconstructed slice remains a valid arbitrage-clean smile. Note 14 develops the directed Bayesian solver behind the propagation (with an optional optimal-transport regularization, off by default: `graphLambdaScale = 0`) and the leave-one-out backtest that measures its skill.

The propagation operator is versioned and adjudicated, and its history is a small case study in how defaults are set here. `precision_messages` — the options default since 2026-07-27 — replaces the smoothness field by pairwise *relation factors* (calendar ladders, index→name pairs) whose per-edge precisions are first-class policy objects with their own editor, live preflight and draft/active lifecycle; its golden acceptance contracts — full transmission, competing signals, dead-informer zero dilution, baseline-uncertainty-enters-once — are locked three times over, against a brute-force Gaussian reference, through the production assembly, and through the HTTP API. Its own pre-registered *daily* gate did not clear (RMS tie, narrow bands, recorded verbatim); the default moved on the intraday async replay, where the smoothness field measures nearly inert while the messages carry the signal — and `smooth_field`, the increment prior of Note 14 and the solver the backtests validated, stays as the wire default, the byte-identity anchor and the explicit rollback. The layered dynamic-harmonic pipeline adds temporal, directed and harmonic structure on top of the messages; its pre-registered daily A/B campaign returned a *negative* verdict at that horizon (residual memory monotone against the half-life, optimum $H \rightarrow 0$), and the intraday replay then *validated* the memory mechanism (interior optimum near a tenth of a day, OU-shaped decay) while measuring its directed spatial carrier behind the messages on the ETF triangle — so it stays opt-in, both verdicts recorded, not overwritten, an instance of the “measure first” rule of section 6. Even the in-app autotune is

the production leave-one-out harness, scoring candidate reach η on real hold-outs rather than on a synthetic sandbox.

Two scoring conventions, used everywhere the graph is evaluated: *skill* is the reduction in held-out ATM error versus the transported-prior baseline, in ATM vol bps — positive means the graph beat mechanical transport; ζ is the standardized held-out residual (realized error divided by the posterior credible band), so $\text{std } \zeta = 1$ means the bands are honest, > 1 overconfident (too narrow), < 1 conservative. Skill is measured at scale: on the 25-asset benchmark across three historical regimes ($\sim 47\text{k}$ held-out scores), neighbour-supported indexes gain $+10$ to $+76$ ATM vol bps over the transported-prior baseline and ETFs $+3$ to $+7$; fully-dark single names behind lit indexes/ETFs — the product case — gain $+7.9$ to $+14.2$ bps in the August-2024 spike and $+3.8$ to $+7.2$ fully out-of-sample in the October-2022 bear, with unbiased standardized residuals. In the calm July-2023 regime the dark-name skill is ≈ 0 (never negative): single-name moves there are earnings-idiosyncratic, and the graph carries shared repricing, not idiosyncratic news. The one weakness that measurement flagged — calm-regime dark-name bands at $\text{std } \zeta \approx 1.9$, overconfident — is now closed for the ATM level: an *idiosyncratic band floor* (a strictly causal floor built from the node’s own trailing unexplained moves while lit, on by default in production) moved the two calm cells $1.91 \rightarrow 1.02$ and $1.85 \rightarrow 1.03$ with the posterior mean untouched. Widening the skew/curvature bands in idiosyncratic tape remains open (Note 14).

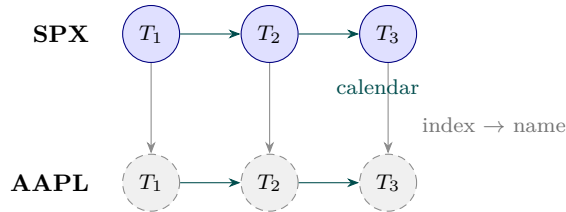


Figure 3: A fragment of the smile graph: lit index nodes (solid) inform dark single-name nodes (dashed) across both calendar and cross-asset edges. The graph posterior reconstructs the dark smiles from the lit innovation.

5 Capability catalogue

Table 1 indexes what the app can do against the note that documents the mathematics. It doubles as a reading order.

Table 1: Capability $\rightarrow$ note index.

Capability	Summary	Note
LQD smile model	Arbitrage-free-by-construction log-quantile density; analytic Jacobian	01
SVI / SVI-JW	Raw and jump-wing SVI; structural chart, buffered Lee cap, belly certificate + one-shot repair	02
Multi-Core Sigmoid (MCS)	Sigmoid base + zero-wing hats for WW/event smiles	03
Local volatility	Jointly calibrated piecewise-affine local- <i>variance</i> surface (Dupire); GN/Numba calibration	04

Capability	Summary	Note
De-Americanization	CRR tree + bisection, Numba kernel, cache digest, wing convex repair	05
Forwards / dividends	Parity forward + discount clamp, discrete divs	06
Weighting & bid-ask	Equal / time-value-density weights; band/haircut fit	07
Variance-swap targets	Replication, LQD closed form, source-PDE	08
Wings & Lee bounds	Tail control unified across models	09
Calendar arbitrage	Convex-order soft hinge, model-agnostic	10
Event variance clock	Dual calendar/variance time, day-weighted events	11
Spot-vol dynamics	Sticky strike/moneyness/local-vol, SSR transport	12
Prior persistence	7-mode Bayesian menu, precision gate, two-pass	13
Graph extrapolation	Transported prior + directed increments; opt-in precision messages + layered dynamics; LOO back-test	14
Observation Kalman filter	Per-handle temporal denoising of the ATM handles; one-stage MAP active mode	15

Viewer and data capabilities (engineering, not derived here). The workspace is organized into eight tabs — Parametric, Local Vol, Forwards, Options, Graph, Quality, Universe and View: charts of prior/current fit vs. quote bands in normalized or fixed strike, quantile and density functions, term structure and event-dilated calendar in vol and variance, var-swap levels, interactive quote select/erase/amend, universe selection, lit/dark node maps, and the graph editor. The Quality tab aggregates publish-readiness per node (fit quality, arbitrage diagnostics — including the belly certificate — and data-age staleness) and drives the export/reporting path; the same gates feed the client-facing certification report (section 6). Data plumbing (providers — including Massive intraday history via flat files where entitled — as-of replay, dividend editor, persistence of named universes and fit history) is described in the project README rather than in this mathematical series.

Remark 1 (Current scope and limits). What the application is *not*, as of this writing. The live universe is desk-scale: validation ran on a 25-asset, three-regime benchmark, and the graph solver is dense — comfortable to a few thousand nodes; a sparse solver is deferred until a universe needs it. The server is a single-process application state, deployed hosted single-tenant first; market-data entitlements are bring-your-own. Sub-daily expiries (0DTE) are research/replay-grade — their exit gates (bitwise replay, hard publish block, warm-slice latency) are certification-locked, but live 0DTE is a post-hosting, client-entitled feature. The app fits, extrapolates and publishes vanilla implied-vol surfaces: it does not price or risk-manage exotics, route orders, or forecast — dividends, borrow and forwards are inputs or option-implied estimates, never predictions.

6 The design rules that recur

Reading the series end to end, a handful of design rules appear again and again — each learned from a production incident, each now documented as a *case file* in its note. They are worth stating once as the app’s engineering philosophy.

1. **The confinement principle** (Note 09). A constraint comparing *two* curves is sampled only where data pins both; a constraint intrinsic to *one* curve holds everywhere, wings included. Four instances: the calendar floor (confined, Note 10), the LV convex wing (confined, Note 04), the MCS Durrleman penalty (extended, Note 03), the de-Am convexity repair (extended, but its *authority* confined to the quoted band, Note 05).
2. **Trust the well-identified parameter.** The parity forward keeps the price level and refuses the noisy slope (Note 06); the prior gate trusts the market exactly where quotes identify a feature and yesterday exactly where they do not (Note 13).
3. **Persist shape, not level.** Operators and factors are level-invariant baskets, so an overnight jump is never damped (Note 13); the graph propagates *increments*, so “no change” is the default dark-node answer (Note 14).
4. **Route each model through its native object.** The var-swap is a closed form for LQD, arithmetic for SVI/MCS, a source PDE for LV (Note 08); the calendar constraint is the asset share for LQD and a total-variance hinge for the overlays (Note 10). Uniform implementations across non-uniform models are a trap.
5. **Measure first, fix second.** The short-dated LV failure was convicted by per-expiry counts before any model change (Note 04); the MCS wing penalty sits where the arb census located the violations (Notes 03, 09); the temporal backtests score priors, the graph and the observation filter against honest held-out baselines (Notes 13–15).

Table 2 indexes the production incidents documented across the series — the fastest way to learn the system is arguably to read these first. The index is not only documentation. Incidents marked † are registered, essentially verbatim, as named cases of the *certification pack* (`backend/backtest/certification.py`): 22 cases across three dimensions — market regimes, data failures, model stress — each pointing at the pytest locks that regression-guard it forever, so validation and production share one definition instead of drifting apart. One command (`python -m backtest.certification run`) re-executes every lock and emits a client-facing HTML report. In this series a case file is an *executable* object: the fastest way to learn the system is to read the incidents, and the fastest way to trust it is to run them.

Table 2: Case-file index: production incidents and their lessons.

Incident	Lesson	Note
The Lee cap that sat exactly on the broken boundary ($\beta = 2$ passes the floor and wing screens with zero penalty yet $g(10) = -0.0485$; a live SPY wing sat at 2.0000 — cap now 1.95) †	a fence must not end on the broken boundary	02, 09
The certified-looking slice with a negative belly (Axel Vogt’s classical parameters pass every wing screen; $\min g \approx -0.033$ mid-range — now the 801-point certificate gates publish) †	screens fence the wings; certify the belly	02
The six-day weekly that broke the LV strike grid (fit RMS 108 $\rightarrow$ 23.5 vol bp after the fix) †	measure first; per-expiry coverage floors	04

Incident	Lesson	Note
The convex wing that flattened SPY (fit RMS 25.7 $\rightarrow$ 2.6 vol bp once the constraint left the quoted range) $\dagger$	constraints stay off quoted territory	04
The reverted global de-Am repair (ATM smile gap) $\dagger$	repairs need confined authority	05
The delayed feed that gapped the ATM smile (discount clamp; failed rate-anchor) $\dagger$	trust the well-identified parameter	06
The fit that took minutes (var-swap replication in the Jacobian)	native-object routing	08
The MCS that invented a put wing (R3 $\times$ R6 ablation, in-sample RMS: input repair alone 92 $\rightarrow$ 25 vol bp, output penalty alone 749, both together 225 with the violation gone)	input repair and output penalty compose	09, 03
The phantom calendar (NVDA/SPY flattened; the 151-vol-bp flattening reproduced, then $\rightarrow$ 0 under confinement) $\dagger$	the confinement principle	10, 09
The jump the strike anchor damped (operators persist shape)	persist shape, not level	13
The backtest that confirmed the hybrid default ($\sim$ 32 vol bp better on the held-out wing, 1117 nodes)	defaults are chosen by held-out evidence	13
The transient-name topology defect (cross-asset skill exactly 0.000 bp) $\dagger$	exact zeros are disconnection, not damping	14
The chart benchmark whose headline favoured the wrong arm (the raw SVI chart's lower arb rate was a survivorship artifact of its non-converged third; conditioned on convergence, the structural chart won every regime median)	condition on convergence before comparing arms	02
The A/B campaign whose arms were byte-identical (a config-hash change silently purged the residual store on every pair, so all four variants collapsed to the control)	an ablation must first prove its arms differ	14
The close-strike contradiction vs. the true gap (per-handle Kalman gains level/skew/curvature = 0.80/0.72/0.03: high where quotes contradict, near zero where they are simply absent)	noise is not a gap: filter and prior split the work	15
The off-diagonal blow-up on coarse chains (cross-handle updates moved ATM vol by 3–28 vol <i>points</i>)	per-handle updates; caps do not police cross-gains	15

7 How the notes are organized

The topic notes, 01–15, are standalone and follow a common skeleton, codified in the series style guide (`Docs/notes/STYLE_GUIDE.md`): an opening that states the *production problem* and the *invariants* being protected (the amber box near the top of every note), a theorem-shaped body

whose central object gets the note’s single boxed equation, a worked example or *case file* told as a production incident, a *traceability table* tying each claim to its module and locking test, and three appendices — **A, a hyperparameter atlas** (every knob, surfaced or hidden, with its default), **B, performance** (measured optimizations, including the rejected ones), and **C, a reference implementation** (≤ 50 lines, executed against the production module before being committed). Figures run through the shared `figures/style.py` and their captions state the lesson, not the axes. Notes cross-reference each other (this note’s table 1 is the index) but never depend on one another’s internals. Numbers and plots are *measured*, not copied: where a fresh run differs from an older design note, the fresh run wins. Every topic also carries one or more standalone *lecture editions* — tellings from a fresh angle (Note 01’s percentile ruler, Note 02’s two languages and its moment map, Note 14’s three priors for a dark universe, ...) — written to the register codified in `Docs/notes/LECTURE_REWRITE_GUIDE.md`. As of 2026-07-27 the lecture editions are the *prevailing* technical notes — audited against the code to the same standard — and the numbered series, this overview included, is kept as reference. This overview is the deliberate exception to the skeleton — it is a map, not a theorem, so it carries no boxed central equation, no traceability table and no performance or reference-implementation appendix; its single appendix is the control index below.

A Principal calibration controls

A *curated* index of the principal tunable parameters: surfaced (the user-facing `FitSettings` and `OptionsSettings`) and hidden (internal constants). It is deliberately not exhaustive — the single source of truth is the Pydantic schema pair in `volfit/api/schemas.py`, whose field docstrings state each knob’s unit, activation condition and effect, plus each topic note’s Appendix A. (Workflow and data-fetch controls — spot polling, chain refetch cadence, streaming refit — live in the same schema and are omitted here as non-calibration knobs.) Each row points to the note that derives its role; defaults are the shipped values. This is a directory, not a derivation — consult the referenced note for meaning.

Table 3: Surfaced parameters (principal; the schema is exhaustive).

Parameter	Default	Role	Note
<code>model</code>	<code>lqd</code>	smile family: <code>lqd</code> / <code>svi</code> / <code>sigmoid</code> (MCS)	01–03
<code>nOrder</code>	6	LQD Legendre order	01
<code>regLambda</code> / <code>regPower</code>	10^{-6} / 1	LQD high-order ridge	01
<code>barrierCenter</code> / <code>barrierScale</code>	0.90 / 50	LQD A_R barrier	01
<code>nCores</code>	2	MCS hat count (schema-clamped at 2)	03
<code>sigmoidRidge</code>	10^{-2}	MCS hat-amplitude ridge	03
<code>sivWingPenaltyPct</code>	100	MCS put-wing Durrleman penalty (0 = off)	03
<code>sviPenaltyWeight</code>	10^3	SVI no-arbitrage penalty	02
<code>leeSlopeMax</code>	1.95	SVI Lee wing-slope cap, strictly buffered under Lee’s bound (2.0 = the broken boundary, explicit config only)	02, 09
<code>sviChart</code>	<code>structural</code>	SVI optimization chart (every iterate fence-clean); <code>raw</code> = historical rollback	02
<code>bellyRepair</code>	on	one-shot belly-hinge repair refit when the certificate fails (kept only if it certifies)	02

Parameter	Default	Role	Note
<code>fitMode</code>	<code>mid</code>	mid / bidask / haircut	07
<code>weightScheme</code>	<code>equal</code>	equal / tv-density weights	07
<code>haircut</code>	0.005	band haircut (vol)	07
<code>midAnchorWeight</code>	0.05	band mid anchor	07
<code>varSwapEnabled</code>	<code>on</code>	var-swap quotes surfaced + penalized	08
<code>varSwapWeightPct</code>	10%	var-swap budget	08
<code>varSwapMethod</code>	<code>static</code>	LV model var-swap: strike replication / source PDE	08
<code>jointCarry / jointCarryEngageBp</code>	<code>off / 25</code>	joint borrow/de-Am fixed-point forwards, engaged per expiry above the borrow gate (below it the parity forward is kept exactly)	05, 06
<code>enforceCalendar</code>	<code>on</code>	ascending- T calendar coupling of Calibrate	10
<code>surfaceSolver</code>	<code>symmetric</code>	calendar coupling: screen + joint GN repair of violating runs (<code>sequential</code> = legacy one-sided floor)	10
<code>calendarWeight</code>	10^6	calendar soft-constraint penalty	10
<code>extrapEnforce</code>	<code>off</code>	tapered no-arb enforcement in the extrapolated region (advisory measurement always on)	09, 10
<code>eventsEnabled</code>	<code>on</code>	event variance-clock master switch	11
<code>normalizeEvents</code>	<code>off</code>	rescale so the 1Y day-weight budget stays 365	11
<code>gridStrikeMode</code>	<code>delta</code>	LV strike-vertex placement (<code>delta</code> / <code>linear</code>)	04
<code>gridXNodes</code>	12	LV strike vertices — a <i>floor</i> in delta mode, exact only in linear mode	04
<code>gridXMinPerExpiry</code>	8	LV short-dated coverage floor	04
<code>gridTNodes</code>	10	<i>floor</i> on positive LV time vertices (the base set — 0, pre-front node, every lit expiry — is always kept)	04
<code>gridRegLambda / gridRegRho</code>	$10^{-2} / 1$	LV roughness penalty / ratio	04
<code>convexWingWeight</code>	10^3	LV wing-convexity penalty (<code>convexWing</code> off by default; tail-confined)	04
<code>frontTieWeight</code>	10^{-2}	ties the free $t = 0$ row to the first data-identified row	04
<code>lvVolCapMult</code>	3.0	LV local-vol cap multiple	04
<code>leftWingSlopeMult</code>	1.5	LV left-wing extrapolation slope multiple	04
<code>timeScheme</code>	<code>implicit</code>	LV PDE time scheme (<code>rannacher</code> opt-in)	04
<code>lvSolver</code>	<code>gn</code>	LV solver: matrix-free GN / scipy trf	04
<code>lvFastKernel</code>	<code>on</code>	Numba vectorized-Thomas Dupire march	04
<code>lvEarlyStop</code>	<code>on</code>	LV cold-fit early stop at the RMS stall	04
<code>localVolEnabled</code>	<code>on</code>	LV calibration + workspace master switch	04
<code>dynamicsRegime</code>	<code>sticky_strike</code>	spot-vol regime seed	12
<code>ssr</code>	2.0	SSR value of the <code>custom</code> regime (inert otherwise)	12
<code>priorPersistenceMode</code>	<code>hybrid</code>	prior mode selector	13

Parameter	Default	Role	Note
priorOperatorSet	ATM, RR25, operators to persist BF25, Var-Swap		13
priorOperatorStrengthPct	50%	operator prior budget	13
priorOperatorRequiredPrecision	1.0	operator activation gate	13
priorOperatorGapExponent	1.0	gate exponent γ	13
priorOperatorBandwidth	0.06	operator leg KDE bandwidth	13
priorOperatorCovarianceMode	diagonal	operator covariance route (full = Jacobian-propagated)	13
priorDataOnlyPrepass	off	two-pass activation (measure precision data-only, then refit)	13
priorFactorSet	ATM, skew, smile factors to persist curvature, VarSwap		13
priorFactorStrengthPct	50%	factor prior budget	13
priorTailAnchorStrengthPct	20%	hybrid tail-anchor budget	13
priorAnchorWeightPct	50%	strike-gap prior budget	13
graphKappaScale κ	1.0	graph temporal precision	14
graphEtaScale η	1.0	graph directed smoothness	14
graphLambdaScale λ	0.0	graph OT regularization (0 = OT term off)	14
graphNu ν	0.1	graph UOT source allowance	14
graphPropagationMode	precision_messages	propagation-operator seed (flipped 2026-07-27 on the intraday replay evidence; <code>smooth_field</code> = explicit rollback, <code>hybrid</code> config-only; the layered dynamic-harmonic pipeline is a per-request opt-in)	14
idioFloor	on	dark-node idiosyncratic credible-band floor	14
functionalBand	on	full 3-handle posterior covariance pushed through the slice map: smile band, var-swap and tail-mass sds (band-only)	14
observationFilterMode	off	filter off / overlay / active (MAP)	15
filterCovarianceMode	jacobian	measurement-covariance route	15
filterProcessVolBpSqrtDay	30	ATM clock process noise (bp/ $\sqrt{\text{day}}$)	15
filterProcessSkewSqrtDay	0.02	skew clock process noise	15
filterProcessCurvSqrtDay	0.05	curvature clock process noise	15
filterTransportNoiseScale	0.10	process std per unit transport distance	15
filterResidualInflation	on	contradiction reads as measurement noise	15
filterAdaptiveSigma	3	innovation-gated adaptive Q (0 = off)	15
filterMaxGain	1.0	per-handle own-gain safety cap	15
filterResetHours	96	maximum data gap predicted across	15
filterClock	calendar	filter time base; <code>session</code> = intraday session clock (a session carries 0.60 of a day's variance, non-trading days 0)	15
filterDataOnlyPrepass	off	strictly persistence-free measurement	15
autoCalibrate	on (gated live server: off)	fetch→fit workflow gate; off = fits wait for explicit Calibrate	—

Per-ticker *market state* — the scheduled-event calendar driving the variance clock (Note 11), dividends,

rates, forward policy, var-swap quotes and lit/dark marks — is data, not a setting: it lives in the per-ticker session state, is edited in the workspaces, and bumps only that ticker’s version (section 1, invariant 4).

Table 4: Hidden internal constants (selected).

Constant	Default	Role	Note
Z_MAX / N_POINTS	40 / 8001	LQD quadrature grid	01
OPT_N_POINTS	2001	LQD optimization grid	01
EPS_AR	10^{-6}	LQD $A_R < 1$ buffer	01
_CERT_POINTS / CERT_G_TOL	801 / 10^{-4}	belly-certificate grid and g tolerance	02
DEFAULT_N_K	1201	LV <i>validator/reprice</i> PDE strike nodes (the affine calibration march has its own grid)	04
DEFAULT_DT_MAX	1/400	LV validator/reprice PDE max time step	04
N_RANNACHER	4	LV validator/reprice Rannacher start steps	04
VAR_FLOOR	10^{-6}	Dupire local-variance floor	04
DEFAULT_STEPS	501	CRR scalar tree depth	05
DEFAULT_BATCH_STEPS	192	CRR batch tree depth	05
BATCH_BISECTIONS	24	de-Am bisection sweeps	05
SIGMA_LO / SIGMA_HI	10^{-4} / 4	implied-vol bracket	05
VS_HALF_WIDTH / VS_POINTS	6 / 801	var-swap replication grid	08
CAL_STRIDE	25	calendar constraint stride	10
DAYS_PER_YEAR	365	calendar→variance clock	11
SPREAD_HALF	0.05	precision spread half-width	13
OBS_FRESHNESS_HALFLIFE	3 d	observation freshness decay	13
TRANSPORT_SCALE	0.10	prior transport precision decay	13
HANDLE_CONFIDENCE	[1, 1, 0.01]	per-handle graph confidence	14
SOURCE_BASE	tiered	prior provenance precision	14
IDIO_FLOOR_LAMBDA	0.30	idio band-floor strength ($\text{sd}^2 \geq \lambda \sigma_I^2$)	14
IDIO_EWMA_HALFLIFE	5 d	idio innovation-RMS half-life	14
DIAGONAL_UPDATE	on	per-handle scalar Kalman gains	15
RESID_INFLATION_CAP	25	cap on the contradiction inflation	15

Reading the index. A surfaced parameter is a *policy* choice — how strong a penalty, which prior mode, how to weight quotes — safe to tune. A hidden constant is tied to an algorithm’s correctness (the A_R buffer, the PDE floor, the tree depth) and is exposed only in the relevant note’s Appendix A, where the consequences of changing it are spelled out. Together with those note-level appendices this index maps the fitter’s control surface; the schema modules in `volfit/api/` remain the exhaustive, always-current record.